

Impact of AI on Advancing Women's Safety

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To the Almighty, who has supported us with steadfast love and support, our parents, family members, loved ones, mentors, instructors, and moral supporters. For all of you, we dedicate this. Your unwavering affection, acceptance of our promises, and faith in our talents have motivated our efforts.

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Hitesh Gehani, Yeshwantrao Chavan College of Engineering, India

Shubhangi Rathkanthiwar, Yeshwantrao Chavan College of Engineering, India

Due to the expanding requirement for security, intelligence-based savvy observation was taken into consideration. Observation frameworks are outlined to detect videos, pictures, sound, or any sort of reconnaissance information without human intercession. Later progress in computer vision, mechanical autonomy, and machine learning have played a noteworthy part in the selection of shrewd frameworks. The framework proposed here will automatically react based on what comes about amid preparation in different modes. This chapter aims to supply an outline of AI-based insights and their capabilities. This chapter also incorporates data around most forms of intelligence-based surveillance frameworks such as following, location, and dissemination of objects and perception behavior.

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(Deemed), India*

The statistics mention that women harassment compared to the male counterparts is significantly higher all over the world. Even though the government has brought in very stringent laws for violence against women, women's safety has been a debatable and major social issue. We are in the era of Digital transformation wherein all our processes and activities are enabled through technology platforms; it is also important to implement the same and enable the women's community to make use of these platforms to make them more resilient and stronger. The technology implementation in women's safety can be categorized into physical, mental, and emotional aspects. AI-based solutions, AI-based collaborative security framework, and the process of AI-enabled security solutions to ensure the holistic security aspects of women are presented here. The increased technology implementation and the great penetration power of technology, right from rural to urban areas and from illiterate to tech-savvy communities has improved the well-being

of women tremendously in the recent past.

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Anita Chaudhary, Akal College of Engineering and Technology, India

Shivani, Akal College of Economics, Commerce, and Management, India

Online harassment and cyber bullying have become serious social issues with wide-ranging repercussions in an age where the internet plays a significant role in our everyday lives. This chapter of the book examines the creation and application of AI-based systems for stopping online bullying and harassment as cutting-edge solutions to this expanding issue. The chapter starts out by looking at how ubiquitous online harassment is and the significant effects it has on both individuals and communities. It explores the difficulties of preventing such behaviours in the digital sphere and draws attention to the shortcomings of conventional moderation techniques. The chapter then delves deeply into the role that AI-based solutions play in content analysis, user profiling, real-time monitoring, and alert production. Additionally, it addresses the crucial issues of ethics and privacy, highlighting the necessity of responsible AI research and data protection.

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In today's technology-driven era, workplace safety remains a paramount global concern. To proactively prevent accidents, mitigate risks, and ensure employee well-being, this abstract introduces the research project 'AI-Driven Predictive Safety Analytics Enhancing Workplace Security.' This initiative leverages artificial intelligence (AI) and data analytics to transform occupational safety. By harnessing historical incident data, real-time monitoring, and advanced machine learning, it aims to create a predictive safety system that identifies and pre-empt potential hazards. Anticipated outcomes include a more secure work environment, reduced accidents, improved well-being, and enhanced efficiency. Empowering decision-makers with actionable insights, this approach enables data-driven, proactive choices, setting the stage for a safer workplace future through cutting-edge technology and data-driven insights.

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With increase in the technology there is increase in the crimes against girls and women. No matter how much Bharat is growing in terms of economy, there is vast growth in sexual assault. So, there is a necessity to analyse this aspect also. The crime against girls and women reported from 2017 to 2021 for various states of Bharat country has been retrieved from the government port and used for the analyses of this research chapter. The data is represented keeping view in every criterion which includes rape, kidnapping, abduction, death due to dowry, insult, and many more. Graphical representation of the data becomes easy to under real time situation of the country. Using this chapter, the authors want to attract the Government of Bharat to understand the situation in the country and take appropriate action in this field.

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Large language models are sophisticated AI systems that process and produce text that resembles that of a human being by learning patterns from enormous volumes of data. These systems make it possible to do jobs like dialogue, translation, and content production. The most popular implementation of LLM is the generated pretrained transformer (GPT) family of LLM. LLMs can assist in women's safety in numerous ways. They can help in distress and prove to be a dependable tool for women in the event of a crisis. LLMs can be employed to create virtual support groups for susceptible women, virtual assistance in times of crisis or need, and enhance the safety of women by suggesting avoidance of certain roads or vicinities for safety concerns. IIm apps can function as safety alarms, safety education programmes, generative AIs to create appealing social media postings, blog entries, and videos, and safety-focused wearable technology.

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Anushka Aggarwal, Asian Business School, India

Shubhika Gaur, Asian Business School, India

The advent of technological advancements has propelled global progress to unprecedented levels; nevertheless, it has also emerged as a potent instrument that can be wielded to compromise the safety of women. These issues have a detrimental effect on the overall welfare and online encounters of individuals. The field of artificial intelligence (AI) has experienced significant growth and has become a formidable resource for tackling the urgent challenges related to women's harassment and cyberbullying. In order to address these challenges, scholars and experts have resorted to employing AI techniques

as proactive strategies for both prevention and intervention. This chapter explores the utilization of artificial intelligence technology in mitigating instances of women's harassment and cyberbullying by the automated identification and response to harmful information, hence promoting the establishment of more secure online spaces.

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In today's society, women's safety and empowerment are top priorities. Artificial intelligence (AI) integration offers a revolutionary means of resolving these problems. This abstract examines a clever and empowering strategy that makes use of AI technologies to improve the safety of women. AI-powered personal safety applications dramatically improve individual security by providing real-time location monitoring, emergency notifications, and connectivity with trusted contacts. The use of AI algorithms in predictive policing detects high-risk regions and patterns of violence against women, allowing for tailored law enforcement responses. AI-enabled safety chatbots and hotlines offer a secure environment for reporting occurrences and provide details on one's legal rights and available assistance options. Platforms for reporting and crowdsourcing data enable women to contribute to data-driven safety efforts, enabling more efficient responses. Initiatives for community interaction powered by AI raise awareness and enable quick solutions to safety issues.

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Maintaining women's security and happiness is critical in today's world. The plan aims to provide women with a comprehensive collection of resources that will increase their personal safety and allow them to deal with daily life more freely. The main features of the app include emergency notifications, real-time location sharing, and virtual escort capabilities. Users can rapidly send distress messages to their pre-designated emergency contacts, including their exact location, to ensure a timely response in critical situations. Furthermore, the software allows users to communicate their present positions with trusted close companions, increasing personal safety when travelling alone. They have very little societal involvement and are ineffectual if it involves ensuring the security of women.

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Enhancing Human Safety and Well-Being: A Smart Wearable Compatible With Comprehensive Android Application for Emergency Response 160

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Risks to one's health can arise for both corporate employees and individuals working in mines, factories, or construction sites. On occasion, they must deal with dangerous circumstances like fire, water, rockfall, poisonous chemical releases at work, poor ventilation, etc. Several workers thus experience breathing difficulties, heart attacks, hypertension, etc. In addition, women who work face the risk of harassment, theft, etc. Even those who unavoidably suffer accidents while away from home for work or any other reason regularly fail to notify their families or have delays in seeking medical care. In this study, an Android app and a smart wristband are proposed which will measure wearer's body temperature, heart rate, pulse, and levels of oxygen saturation and send the data to the Android application on the user's mobile device with the 4G long term evolution (4G LTE) module and the internet of things (IoT) framework. This app will even update the user's family about mishappenings.

Chapter 11

Inferring Personality From Social Media User Behaviors Using Dense Net Convolutional Neural Networks 173

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We live in a world where social media is omnipresent and integrated into our daily lives. People love to express their interests, thoughts, and opinions on these social networking platforms. This information reveals several psychological aspects of their behavior and can be used to predict their personality. To predict this, introduce the method dense net convolutional neural network (DNCNN) is based on predicting the social media users' personality identification. Performed an experimental evaluation on a benchmark dataset for the task of categorizing personality traits into distinct classifications. The review of the dataset yields improved results, showing that the proposed model can really arrange client character attributes when contrasted with cutting-edge models. Posts and status updates can be used to predict the personality of users of social media networks to improve accuracy. These results show that picture features are better predictors of personality than text features, and also found that a profile picture reliably predicts personality with 96% accuracy.

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IoT-Based Smart Safety Analyzer for Women 185

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Despite significant technological advancements in modern times, the safety of women remains a persistent concern. It's important to develop IoT-based women's security systems with a user-centric approach, involving women in the design process to ensure the technology meets their needs effectively. An internet of things (IoT) based women's security system leverages connected devices and sensors to enhance the safety and well-being of women. These systems use technology to monitor, communicate, and respond to potential threats or emergencies. Artificial intelligence-based tools are employed to distinguish male individuals within designated women-only areas. The system assesses conditions by comparing specific attributes against its internal safety database. Upon detecting a potentially concerning scenario, the system promptly notifies the staff and triggers audible alarms for immediate attention. A "smart safety analyzer for women" could refer to a technological solution aimed at enhancing the safety and security of women in metro train.

Chapter 13

Mobile Application-Based Women's Safety and Security System Using AI 200

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There are many shameless residential manhandles for ladies across the world. Usually quickened due to the nonappearance of a successful following framework. This is centred on a female security framework centred on AI which offers security to women. This system can be programmed and manually react accurately in pivotal circumstances. In arrange to accurately track the condition of casualty and Raspberry Pi, the proposed gadget comprises a discourse acknowledgment device, Gsm modem, and a few other locators, such as beat locators and an accelerometer, to screen the information and shirking or security measures might be carried out agreeing to the noteworthiness of the issue. The authors utilize one program and a few collar chain artifacts and shows that are in regular utilize. The machine is indistinguishable from a screen on the collar with a button as a source, where, when turning on the yelling warning, the electric stun gadget the screen and area points of interest of contacts and the closest police station are implemented for the self-defense aim.

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Mobile Apps for Personal Safety of Women Using Blockchain Technology 216

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The safety and security of women in today's society is of paramount concern, given the myriad of challenges and threats they face. In response to these pressing issues, this chapter explores the convergence of technology and social welfare through the lens of "Mobile Apps Using Blockchain for Women's Safety." The chapter commences by illuminating the gravity of the problem, emphasizing the need for innovative solutions. It delves into the fundamental aspects of blockchain technology, elucidating its decentralized, transparent, and secure nature, highlighting why it is aptly suited to address women's safety concerns.

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Unlocking Insights: Ethical Considerations and Classifications of Data Analytics for Social Networks 238

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The social media analytics, which relate to huge amount of data from various social media platforms, are used to understand an opinion from the written language such as tweets, chats, comments. In existing methods, the sentiment analysis on Xcorp (Twitter) was mostly used for emotion detection for the polling methods; and star ratings are used to see the response from people. In this model, using Twitter API to fetch the data and Naive Bayes model for classifying them. The tweets, retweets and comments are collected and processed with the positive, neutral, and negative responses from the user will be reflected for ethical considerations. The information obtained from this system is used in various applications like analysis of social media support for politicians, safety technology in social media, a review based on user response for a product, response from people or government for social or political issues (Hashtags), movies, etc. The system will help to visualize statistics to analyse people's responses to provide the most effective statistical tool for various industries.

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Voice-Activated SOS: An AI-Enabled Wearable Device	251
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In an era marked by technological innovation, the convergence of artificial intelligence (AI) and wearable technology has yielded transformative solutions. Among these, the Voice-Activated SOS feature stands as a pioneering advancement, revolutionizing personal safety by harnessing the power of AI to respond swiftly to distress calls. This chapter provides a comprehensive exploration of this groundbreaking technology, offering insights into its development, functionality, and its profound impact on personal security. At the heart of the Voice-Activated SOS feature lies advanced voice recognition AI, capable of distinguishing distress signals from regular speech. This technology, seamlessly integrated into wearable devices, empowers users to activate a distress call with a simple voice command. Once activated, the device initiates a rapid communication workflow, transmitting alerts to designated emergency contacts along with real-time location data.

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Women are facing different problems in different stages. The importance of comprehending women's personal and health issues is highlighted in the opening section. Women are suffering from so many problems, such as peer pressure, depression and anxiety, financial issues, transportation blockades, time management, girls' security and safety, homesickness, career choice perspective, early marriage issues, physical activity and nutrition, substance abuse, mental health issues, physical injury, and violence. The data is grouped by age; for instance, age 8 to 12, age 12 to 16, age 17 to 24, age 25 to 30, age 30 to 40, age 40 to 60, and age 60 to 80. Using deep learning techniques, the proposed system aims to conceptualize and understand these issues, and to develop one website which would provide a solution for each and every query with consulted name and address.

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Foreword

A shining example of technological innovation, artificial intelligence has the potential to alter the fundamental foundations of our civilizations. We must seriously assess the influence of AI on many aspects of our lives, particularly those that have traditionally suffered obstacles and inequities, as we move through this landscape of possibilities. The book *Impact of AI on Advancing Women's Safety* is a riveting voyage into the convergence of technology and gender, illuminating one essential and sometimes ignored facet of the AI revolution.

The right to feel safe is paramount, and for a very long time, women have struggled with concerns that threaten their safety. This book, which explores how artificial intelligence might be used to improve women's safety in today's dynamic society, stands out as an essential and relevant addition to the conversation. The authors go deeply into the complexities of this connection, providing a comprehensive study of the advantages and disadvantages that artificial intelligence may bring.

This collection combines in-depth research, case studies, and the opinions of experts to shed light on the many ways in which artificial intelligence technology might be used to make communities safer for women. The authors provide various AI-powered solutions, from intelligent surveillance systems to new personal safety applications, to address the specific issues women confront in different settings.

This book, however, is not a simple ode to technical progress. It's a rallying cry for making AI's advantages available to everyone and a call to action to ensure that happens. The authors urge us to think about the ethical implications so that we don't unintentionally reinforce existing prejudices or increase gender gaps when we use AI in safety measures. They inspire us to think of a future in which artificial intelligence is used ethically and responsibly to make the world safer and fairer for women.

The concepts in this book serve as a compass as we approach a new age in which artificial intelligence (AI) enters every area of our lives, leading us to a tomorrow in which everyone realizes the benefits of technology. To get closer to a society where women may live without fear, the authors encourage governments, engineers, and campaigners to work together to define the future and harness AI's revolutionary potential.

In the following pages, readers will discover a compelling vision for a future where technology becomes a catalyst for good change and a thorough analysis of the influence of AI on women's safety. This book is a must-read for anybody interested in learning about, contributing to, and shaping the junction of AI and improving women's safety. We hope it will motivate people to work together to use technology to its fullest potential for the common good.

This book deepens into the unique viewpoints and creative strategies for improving women's welfare programs using modern computing capabilities. It is a gathering of thought leaders committed to using technology to improve social welfare systems in terms of efficiency, effectiveness, and fairness.

Foreword

This book's chapters cover various topics and disciplines, including healthcare, education, poverty reduction, social justice, and community development. The authors provide a deep dive into the specifics of optimization methods, data analysis, artificial intelligence, machine learning, and other emerging technologies that potentially transform how we approach social problems.

The authors provide conceptual models, case studies, and real-world applications to back up their claims that technology may revolutionize the field of social welfare. The authors explain how technology progress might be used to create more equitable, sustainable, and empathetic communities by sharing their own experiences and insights.

I sincerely desire that this book be a source of motivation, insight, and contemplation for academics, professionals, and students in various disciplines. The ideas presented here have the potential to influence the future of social welfare, influencing policy choices, directing the creation of novel solutions, and ultimately bettering the lives of people and communities everywhere.

I appreciate the authors' hard work and dedication to moving the discussion of advancing women's safety with AI employing cutting-edge software. I want to thank the book's editors and production staff for their tireless efforts in making this vital collection a reality.

Hopefully, this book will pique readers' interest, encourage them to work together and pave the way toward a future in which technology is used effectively to advance gender equality and human development.

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Preface

As the editors of this timely reference book, *Impact of AI on Advancing Women's Safety*, we are pleased to present a comprehensive exploration of the intersection between technology, artificial intelligence, and the critical issue of women's safety. In recent years, artificial intelligence has opened up new possibilities to enhance women's security in public and private spheres. This edited volume brings together a wealth of insights from experts in the field, shedding light on the potential of AI to make meaningful contributions toward fostering a safer world for women.

AI has ushered in a new era of safety tools designed to empower women in diverse settings. From smart gadgets to personalized applications, these AI-driven solutions offer a range of features, including location monitoring, real-time threat assessment, and emergency response coordination. The ability to tailor these technologies to individual needs marks a significant stride toward providing women with a heightened sense of independence and peace of mind in their daily lives.

In this book, we delve into various dimensions of AI's impact on women's safety, covering topics such as predictive analytics, intelligent surveillance systems, mobile apps, cyberbullying prevention, predictive policing, education, and ethical considerations. Each chapter contributes to a broader understanding of how AI tools can be harnessed to address women's safety concerns while acknowledging their deployment's social, ethical, and legal implications.

The potential of AI to combat sexism, harassment, and violence against women is profound. The applications are vast, from video analytics and facial recognition software monitoring public spaces to AI chatbots providing a secure environment for reporting incidents. The ability of AI algorithms to identify and rectify bias in decision-making processes within criminal justice systems offers a glimmer of hope for a more just and equitable society.

However, with these promises come challenges. The ethical use of data, algorithmic bias, and privacy concerns demand careful consideration. As technology continues to evolve, so must our commitment to using it ethically and ensuring that it serves the interests of all without perpetuating existing inequities.

Chapters within this comprehensive reference book collectively form a compelling narrative that spans the realms of intelligent surveillance, digital transformation, combating online harassment, predictive safety analytics in the workplace, and the socio-technological landscape of crimes against women. From the analysis of government-reported data in Bharat to the innovative applications of Large Language Models and the revolutionary potential of blockchain in women's safety, each chapter contributes unique insights and solutions. Delving into the ethical considerations, privacy concerns, and proactive strategies for women's security, this collection is an indispensable guide for policymakers, law enforcement agencies, and technologists. It advocates dedicated to fostering a world where women can live free from fear and fully participate in all aspects of society. The chapters span a spectrum of AI applications, from

Preface

wearable technology to sentiment analysis on social media, providing a holistic perspective on leveraging cutting-edge technology to address the multifaceted challenges surrounding women's safety.

Chapter 1 provides an in-depth exploration of intelligent surveillance systems designed to enhance women's safety. As the demand for security continues to grow, advancements in computer vision, robotics, and machine learning have paved the way for sophisticated surveillance frameworks. The proposed system responds automatically based on training outcomes, offering various modes to adapt to different scenarios. The chapter offers an overview of AI-based insights and their capabilities, encompassing various forms of intelligent surveillance, including tracking, location monitoring, object distribution, and behaviour perception.

Addressing the significant global issue of women's safety, Chapter 2 examines the implementation of AI-driven solutions to empower women. Despite stringent laws, women's safety remains a contentious social problem. Technology platforms can make women more resilient and vital in the digital transformation era. The chapter categorizes technology implementation into physical, mental, and emotional aspects, presenting AI-based collaborative security frameworks and processes to ensure holistic security. From rural to urban areas, widespread technological penetration has significantly improved women's well-being.

Chapter 3 delves into the pervasive issues of online harassment and cyberbullying, offering cutting-edge AI-based solutions to address these challenges. It highlights the prevalence and effects of online harassment, discussing the limitations of traditional moderation techniques. The chapter explores the role of AI in content analysis, user profiling, real-time monitoring, and alert generation. Emphasis is placed on the ethical considerations and privacy concerns associated with implementing AI solutions, underscoring the importance of responsible AI research and data protection.

Focusing on workplace safety in the technology-driven era, Chapter 4 introduces a research project leveraging AI and data analytics. The initiative aims to proactively prevent accidents, mitigate risks, and enhance employee well-being through a predictive safety system. The chapter anticipates a more secure work environment, reduced accidents, improved well-being, and enhanced efficiency by utilizing historical incident data, real-time monitoring, and advanced machine learning. Decision-makers are empowered with actionable insights, paving the way for a safer workplace future through cutting-edge technology.

Chapter 5 tackles the increase in crimes against women in Bharat, offering a data-driven analysis of reported incidents from 2017 to 2021. The research utilizes government data to represent various criteria such as rape, kidnapping, dowry-related deaths, and more. Graphical representations provide real-time insights into the country's situation, urging the government to understand and appropriately address these challenges.

Exploring the role of Large Language Models (LLMs) in women's safety, Chapter 6 introduces the applications of sophisticated AI systems like GPT in distress situations. LLMs are discussed for creating virtual support groups, offering virtual assistance during crises, and enhancing women's safety through personalized recommendations. The chapter sheds light on how LLMs contribute to safety education, generating content and wearable technology designed to focus on women's safety.

Chapter 7 examines the growth of AI in addressing challenges related to women's harassment and cyberbullying. It explores proactive strategies for prevention and intervention, utilizing AI techniques for automated identification and response to harmful information. The focus is on creating secure online spaces and the ethical considerations for employing AI in these efforts.

Highlighting the priority of women's safety and empowerment, Chapter 8 explores the integration of AI to resolve these challenges. It discusses AI-powered personal safety applications, predictive policing,

Preface

safety chatbots, and reporting platforms. The chapter emphasizes how AI contributes to creating a safer environment for women and suggests avenues for community interaction to raise awareness and prompt solutions to safety issues.

Chapter 9 addresses the critical need for women's security and happiness, introducing AI-enhanced mobile apps as a solution. The features include emergency notifications, real-time location sharing, and virtual escort capabilities. Users can communicate distress messages to designated contacts, enhancing personal safety during travel or in critical situations. The chapter underscores the societal impact of AI in ensuring women's security.

Focusing on health and safety risks faced by women in various workplaces, Chapter 10 proposes an Android app and smart wristband leveraging AI. The system monitors vital signs and sends data to a mobile app, updating the user's family about any mishaps. It addresses issues such as harassment, theft, and accidents, emphasizing the role of AI in providing timely assistance and ensuring women's well-being.

Chapter 11 explores the use of Dense Net Convolutional Neural Network (DNCNN) for predicting social media users' personalities based on their posts and status updates. The model utilizes image features and presents improved results in categorizing personality traits. The chapter highlights the potential of AI in predicting personality traits from social media data, providing valuable insights for various applications.

In response to persistent safety concerns, Chapter 12 introduces IoT-based women's security systems with a user-centric approach. It leverages connected devices and sensors, incorporating AI tools to distinguish individuals within designated women-only areas. The system assesses conditions, triggering alarms and notifying staff in potentially concerning scenarios. The chapter emphasizes a "smart safety analyzer for women" as a technological solution to enhance women's safety.

Addressing the need for effective tracking systems, Chapter 13 introduces a female security system based on AI. The proposed device, utilizing speech recognition, a GSM modem, and various locators, aims to provide accurate responses in crucial situations. The chapter discusses the implementation of electric shock devices and location details for self-defence purposes.

Chapter 14 explores the convergence of technology and social welfare through the lens of mobile apps using blockchain for women's safety. It emphasizes blockchain technology's decentralized, transparent, and secure nature, showcasing its potential to address women's safety concerns through innovative solutions.

Chapter 15 focuses on social media analytics for sentiment analysis, using the Naive Bayes model and Twitter API to fetch and classify data. The information obtained is applied in various applications, including analyzing social media support for politicians, safety technology in social media, and reviews based on user responses for products, social and political issues, movies, etc. The chapter highlights the system's potential as an effective statistical tool for various industries.

Examining the transformative solutions arising from the convergence of AI and wearable technology, Chapter 16 explores the Voice-Activated SOS feature. The feature uses advanced voice recognition AI to distinguish distress signals from regular speech, empowering users to activate a distress call with a simple voice command. The chapter provides insights into this groundbreaking technology's development, functionality, and profound impact on personal security.

Chapter 17 emphasizes the importance of understanding women's personal and health issues across different age groups. The proposed system uses deep learning techniques to conceptualize and address various issues such as peer pressure, depression, financial concerns, safety, and health problems specific to different age brackets. The chapter introduces a web platform as a solution to provide guidance and support for women's diverse challenges.

Preface

This book is not just for scholars and researchers; it is a resource for policymakers, law enforcement agencies, technology developers, women's rights advocates, and anyone committed to creating a safer world for women. We aim to foster informed discussions and inspire further research and innovation in women's safety by exploring the potential advantages and pitfalls of AI and emerging technologies.

As editors, we thank the contributors who have enriched this volume with their expertise and diverse perspectives. We hope this book serves as a valuable resource for understanding, implementing, and advancing the role of AI in promoting women's safety. Together, let us harness the power of technology to create a world where women can live free from fear and fully participate in all facets of society.

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Acknowledgement

Many people need support, direction, and participation in the collaborative process of writing a book. As we complete our work on the *Impact of AI on Advancing Women's Safety*, we sincerely thank everyone who helped make this endeavor possible.

We express our heartfelt gratitude to the Supreme Being, our Parents, and our extended Family for their continuous love, assistance, and counsel throughout our lives. Our appreciation extends to our beloved family members who have stood by us in our professional journeys, contributing to the refinement of this book. The steadfast encouragement, belief in our abilities, and enduring affection you have shown us have served as the bedrock that propelled us forward in this undertaking.

We want to express our sincere gratitude to every author for contributing their insightful opinions, vast experience, and thorough research to this book. Your enthusiasm for social welfare applications and eagerness to impart knowledge have greatly aided in developing a comprehensive and informative resource. It was determined that every chapter in the book was necessary; otherwise, it wouldn't have been complete.

Furthermore, we acknowledge and value the meticulous efforts and precious time invested by every member of our editorial board and chapter reviewers in enhancing the quality of the information within the book. We extend our thanks to the reviewers who diligently scrutinized the chapters, provided constructive criticism, and played a pivotal role in elevating the overall standard of the content. Your expertise and discerning analysis have been instrumental in enhancing the scholarly merit of this book.

We want to thank the IGI Global editorial and production teams for their hard work in making this book a reality. Your dedication to excellence, professionalism, and attention to detail have benefitted the entire publishing process.

We appreciate our coworkers' and peers' support as we prepared this book. Your support, conversations, and experiences with us have shaped our viewpoints and improved the information in our work.

We want to express our sincere gratitude to everyone who helped write this book, whether they were directly involved or not. *Impact of AI on Advancing Women's Safety* is the result of our collaborative efforts, and anticipating significant value, we believe that this resource will be instrumental in enhancing women's safety through AI technology.

Chapter 1

AI-Based Smart Surveillance System: A Paradigm Shift in Security for Women and Beyond

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ABSTRACT

Due to the expanding requirement for security, intelligence-based savvy observation was taken into consideration. Observation frameworks are outlined to detect videos, pictures, sound, or any sort of reconnaissance information without human intercession. Later progress in computer vision, mechanical autonomy, and machine learning have played a noteworthy part in the selection of shrewd frameworks. The framework proposed here will automatically react based on what comes about amid preparation in different modes. This chapter aims to supply an outline of AI-based insights and their capabilities. This chapter also incorporates data around most forms of intelligence-based surveillance frameworks such as following, location, and dissemination of objects and perception behavior.

1. INTRODUCTION

Security is getting to be very challenging in present-day time society to halt the issues faced by individuals in their lives conjointly their resources which are profitable from unlawful taking care of. As a result, personal social security and security are important in arranging to secure the personal information of each and each person moreover their day-to-day exercises and important things. A huge sum of closed-circuit security cameras with diverse modules of sensors has been outlined created and deployed to screen the basic foundation offices such as airports, military bases, managing an account, control plants, campuses, etc. The observation of security cameras in manual mode by human administrators isn't an efficient solu-

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Chapter 2

AI-Based Security Framework for Emotional and Personal Safety of Women

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ABSTRACT

The statistics mention that women harassment compared to the male counterparts is significantly higher all over the world. Even though the government has brought in very stringent laws for violence against women, women's safety has been a debatable and major social issue. We are in the era of Digital transformation wherein all our processes and activities are enabled through technology platforms; it is also important to implement the same and enable the women's community to make use of these platforms to make them more resilient and stronger. The technology implementation in women's safety can be categorized into physical, mental, and emotional aspects. AI-based solutions, AI-based collaborative security framework, and the process of AI-enabled security solutions to ensure the holistic security aspects of women are presented here. The increased technology implementation and the great penetration power of technology, right from rural to urban areas and from illiterate to tech-savvy communities has improved the well-being of women tremendously in the recent past.

1. INTRODUCTION

Women safety is always a debatable and complex issue which comprises physical, social and emotional aspects. In the Indian context, Women have been a central force in human history since the Vedic ages. It is clearly depicted that they are always protected by the male community as a father, brother or husband. Even though we portray that woman as a Shakthi Roop, the recent trends show that the atrocities against women have increased multi-fold in the recent past. The problems against women start with

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Chapter 3

AI-Based Online Harassment and Cyber Bullying Prevention System

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ABSTRACT

Online harassment and cyber bullying have become serious social issues with wide-ranging repercussions in an age where the internet plays a significant role in our everyday lives. This chapter of the book examines the creation and application of AI-based systems for stopping online bullying and harassment as cutting-edge solutions to this expanding issue. The chapter starts out by looking at how ubiquitous online harassment is and the significant effects it has on both individuals and communities. It explores the difficulties of preventing such behaviours in the digital sphere and draws attention to the shortcomings of conventional moderation techniques. The chapter then delves deeply into the role that AI-based solutions play in content analysis, user profiling, real-time monitoring, and alert production. Additionally, it addresses the crucial issues of ethics and privacy, highlighting the necessity of responsible AI research and data protection.

INTRODUCTION

The advent of the digital age has brought with it unprecedented opportunities for communication, connection, and collaboration. However, it has also given rise to a more sinister facet of the internet world: cyber bullying and online abuse. These pervasive problems have wide-ranging effects on people, communities, and societies as a whole. Dealing with the problems caused by online harassment has emerged as a top priority as our lives become more and more entwined with the digital world. Online bullying

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Chapter 4

AI–Driven Predictive Safety Analytics: Enhancing Workplace Security

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ABSTRACT

In today's technology-driven era, workplace safety remains a paramount global concern. To proactively prevent accidents, mitigate risks, and ensure employee well-being, this abstract introduces the research project 'AI-Driven Predictive Safety Analytics Enhancing Workplace Security.' This initiative leverages artificial intelligence (AI) and data analytics to transform occupational safety. By harnessing historical incident data, real-time monitoring, and advanced machine learning, it aims to create a predictive safety system that identifies and pre-empts potential hazards. Anticipated outcomes include a more secure work environment, reduced accidents, improved well-being, and enhanced efficiency. Empowering decision-makers with actionable insights, this approach enables data-driven, proactive choices, setting the stage for a safer workplace future through cutting-edge technology and data-driven insights.

1. INTRODUCTION

According to Rasmussen (1998), the fundamental cause of accidents often lies in human errors committed by individuals directly involved in the unfolding events, with statistics indicating that 70-80% of industrial accidents stem from such errors. Human errors are correctable through behavioural interven-

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Chapter 5

Analysis of Crime Against Bharat Girls and Women: Introduction to Women's Safety and Technology

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ABSTRACT

With increase in the technology there is increase in the crimes against girls and women. No matter how much Bharat is growing in terms of economy, there is vast growth in sexual assault. So, there is a necessity to analyse this aspect also. The crime against girls and women reported from 2017 to 2021 for various states of Bharat country has been retrieved from the government port and used for the analyses of this research chapter. The data is represented keeping view in every criterion which includes rape, kidnapping, abduction, death due to dowry, insult, and many more. Graphical representation of the data becomes easy to under real time situation of the country. Using this chapter, the authors want to attract the Government of Bharat to understand the situation in the country and take appropriate action in this field.

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Chapter 6

Application of Large language Models (LLMs) in Women's Safety

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ABSTRACT

Large language models are sophisticated AI systems that process and produce text that resembles that of a human being by learning patterns from enormous volumes of data. These systems make it possible to do jobs like dialogue, translation, and content production. The most popular implementation of LLM is the generated pretrained transformer (GPT) family of LLM. LLMs can assist in women's safety in numerous ways. They can help in distress and prove to be a dependable tool for women in the event of a crisis. LLMs can be employed to create virtual support groups for susceptible women, virtual assistance in times of crisis or need, and enhance the safety of women by suggesting avoidance of certain roads or vicinities for safety concerns. Llm apps can function as safety alarms, safety education programmes, generative AIs to create appealing social media postings, blog entries, and videos, and safety-focused wearable technology.

INTRODUCTION

Women are vulnerable to violence in both public and private settings, including the home and the area around it, as well as in neighbourhoods and cities. Risk is influenced by decisions made about urban design and how public services, such as transportation and electricity, are organized. Greater levels of insecurity are felt by women, which can limit their access to and use of the city (Sankar et al., 2022).

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Chapter 7

Applying Artificial Intelligence to Explore Online Harassment and Cyberbullying Prevention

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ABSTRACT

The advent of technological advancements has propelled global progress to unprecedented levels; nevertheless, it has also emerged as a potent instrument that can be wielded to compromise the safety of women. These issues have a detrimental effect on the overall welfare and online encounters of individuals. The field of artificial intelligence (AI) has experienced significant growth and has become a formidable resource for tackling the urgent challenges related to women's harassment and cyberbullying. In order to address these challenges, scholars and experts have resorted to employing AI techniques as proactive strategies for both prevention and intervention. This chapter explores the utilization of artificial intelligence technology in mitigating instances of women's harassment and cyberbullying by the automated identification and response to harmful information, hence promoting the establishment of more secure online spaces.

1. INTRODUCTION

Artificial intelligence (AI) is an emerging technology that holds significant potential for enhancing worker productivity, improving corporate efficiency, and fostering innovation in the development of novel products and services. Rendering to the Organization for Economic Co-operation and Development (OECD), an artificial intelligence (AI) system is categorized as a mechanism -based system that possesses the skill to impact the surrounding environment by generating an output, such as forecasts, suggestions, or judgments, in alignment with a specified set of objectives. In order to (i) perceive real-

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Chapter 8

Artificial Intelligence: A Smart and Empowering Approach to Women's Safety

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ABSTRACT

In today's society, women's safety and empowerment are top priorities. Artificial intelligence (AI) integration offers a revolutionary means of resolving these problems. This abstract examines a clever and empowering strategy that makes use of AI technologies to improve the safety of women. AI-powered personal safety applications dramatically improve individual security by providing real-time location monitoring, emergency notifications, and connectivity with trusted contacts. The use of AI algorithms in predictive policing detects high-risk regions and patterns of violence against women, allowing for tailored law enforcement responses. AI-enabled safety chatbots and hotlines offer a secure environment for reporting occurrences and provide details on one's legal rights and available assistance options. Platforms for reporting and crowdsourcing data enable women to contribute to data-driven safety efforts, enabling more efficient responses. Initiatives for community interaction powered by AI raise awareness and enable quick solutions to safety issues.

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Chapter 9

Empowering Women's Safety: Strategies, Challenges, and Implications

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ABSTRACT

Maintaining women's security and happiness is critical in today's world. The plan aims to provide women with a comprehensive collection of resources that will increase their personal safety and allow them to deal with daily life more freely. The main features of the app include emergency notifications, real-time location sharing, and virtual escort capabilities. Users can rapidly send distress messages to their pre-designated emergency contacts, including their exact location, to ensure a timely response in critical situations. Furthermore, the software allows users to communicate their present positions with trusted close companions, increasing personal safety when travelling alone. They have very little societal involvement and are ineffectual if it involves ensuring the security of women.

INTRODUCTION

In today's rapidly evolving technological environment, ensuring women's safety and happiness has become a necessity. The proposed plan offers an innovative mobile application that seeks to empower women by providing comprehensive resources, improving personal safety and enabling them to navigate their daily lives with greater freedom and confidence. In this introductory chapter, we will explore the critical importance of women's security and the transformative potential of technology in addressing

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Chapter 10

Enhancing Human Safety and Well-Being: A Smart Wearable Compatible With Comprehensive Android Application for Emergency Response

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ABSTRACT

Risks to one's health can arise for both corporate employees and individuals working in mines, factories, or construction sites. On occasion, they must deal with dangerous circumstances like fire, water, rockfall, poisonous chemical releases at work, poor ventilation, etc. Several workers thus experience breathing difficulties, heart attacks, hypertension, etc. In addition, women who work face the risk of harassment, theft, etc. Even those who unavoidably suffer accidents while away from home for work or any other reason regularly fail to notify their families or have delays in seeking medical care. In this study, an Android app and a smart wristband are proposed which will measure wearer's body temperature, heart rate, pulse, and levels of oxygen saturation and send the data to the Android application on the user's mobile device with the 4G long term evolution (4G LTE) module and the internet of things (IoT) framework. This app will even update the user's family about mishappenings.

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Chapter 11

Inferring Personality From Social Media User Behaviors Using Dense Net Convolutional Neural Networks

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ABSTRACT

We live in a world where social media is omnipresent and integrated into our daily lives. People love to express their interests, thoughts, and opinions on these social networking platforms. This information reveals several psychological aspects of their behavior and can be used to predict their personality. To predict this, introduce the method dense net convolutional neural network (DNCNN) is based on predicting the social media users' personality identification. Performed an experimental evaluation on a benchmark dataset for the task of categorizing personality traits into distinct classifications. The review of the dataset yields improved results, showing that the proposed model can really arrange client character attributes when contrasted with cutting-edge models. Posts and status updates can be used to predict the personality of users of social media networks to improve accuracy. These results show that picture features are better predictors of personality than text features, and also found that a profile picture reliably predicts personality with 96% accuracy.

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Chapter 12

IoT–Based Smart Safety Analyzer for Women

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ABSTRACT

Despite significant technological advancements in modern times, the safety of women remains a persistent concern. It's important to develop IoT-based women's security systems with a user-centric approach, involving women in the design process to ensure the technology meets their needs effectively. An internet of things (IoT) based women's security system leverages connected devices and sensors to enhance the safety and well-being of women. These systems use technology to monitor, communicate, and respond to potential threats or emergencies. Artificial intelligence-based tools are employed to distinguish male individuals within designated women-only areas. The system assesses conditions by comparing specific attributes against its internal safety database. Upon detecting a potentially concerning scenario, the system promptly notifies the staff and triggers audible alarms for immediate attention. A "smart safety analyzer for women" could refer to a technological solution aimed at enhancing the safety and security of women in metro train.

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Chapter 13

Mobile Application– Based Women’s Safety and Security System Using AI

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ABSTRACT

There are many shameless residential manhandles for ladies across the world. Usually quickened due to the nonappearance of a successful following framework. This is centred on a female security framework centred on AI which offers security to women. This system can be programmed and manually react accurately in pivotal circumstances. In arrange to accurately track the condition of casualty and Raspberry Pi, the proposed gadget comprises a discourse acknowledgment device, Gsm modem, and a few other locators, such as beat locators and an accelerometer, to screen the information and shirking or security measures might be carried out agreeing to the noteworthiness of the issue. The authors utilize one program and a few collar chain artifacts and shows that are in regular utilize. The machine is indistinguishable from a screen on the collar with a button as a source, where, when turning on the yelling warning, the electric stun gadget and the screen and area points of interest of contacts and the closest police station are implemented for the self-defense aim.

1. INTRODUCTION

A portable phone with computer highlights is not an exemption. Or maybe it isn’t easy to discover among cutting-edge products a portable phone that’s as it were, a cellular phone without extra choices like capacities to connect with computerized frameworks (PC, portable workstations, etc.) and to get to the Web (Web, mail, etc.). These days versatile innovations are quickly creating. Some individuals

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Chapter 14

Mobile Apps for Personal Safety of Women Using Blockchain Technology

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ABSTRACT

The safety and security of women in today's society is of paramount concern, given the myriad of challenges and threats they face. In response to these pressing issues, this chapter explores the convergence of technology and social welfare through the lens of "Mobile Apps Using Blockchain for Women's Safety." The chapter commences by illuminating the gravity of the problem, emphasizing the need for innovative solutions. It delves into the fundamental aspects of blockchain technology, elucidating its decentralized, transparent, and secure nature, highlighting why it is aptly suited to address women's safety concerns.

1. INTRODUCTION

In an era of remarkable technological advancement, the quest for a safer world remains a compelling and urgent concern. Within this overarching pursuit, the safety and well-being of women have emerged as a critical focal point, deserving both our undivided attention and innovative solutions. The global narrative resounds with the call for women's safety—a call echoed not only by activists but by society as a whole.

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Chapter 15

Unlocking Insights: Ethical Considerations and Classifications of Data Analytics for Social Networks

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ABSTRACT

The social media analytics, which relate to huge amount of data from various social media platforms, are used to understand an opinion from the written language such as tweets, chats, comments. In existing methods, the sentiment analysis on Xcorp (Twitter) was mostly used for emotion detection for the polling methods; and star ratings are used to see the response from people. In this model, using Twitter API to fetch the data and Naive Bayes model for classifying them. The tweets, retweets and comments are collected and processed with the positive, neutral, and negative responses from the user will be reflected for ethical considerations. The information obtained from this system is used in various applications like analysis of social media support for politicians, safety technology in social media, a review based on user response for a product, response from people or government for social or political issues (Hashtags), movies, etc. The system will help to visualize statistics to analyse people's responses to provide the most effective statistical tool for various industries.

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Chapter 16

Voice-Activated SOS: An AI-Enabled Wearable Device

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ABSTRACT

In an era marked by technological innovation, the convergence of artificial intelligence (AI) and wearable technology has yielded transformative solutions. Among these, the Voice-Activated SOS feature stands as a pioneering advancement, revolutionizing personal safety by harnessing the power of AI to respond swiftly to distress calls. This chapter provides a comprehensive exploration of this groundbreaking technology, offering insights into its development, functionality, and its profound impact on personal security. At the heart of the Voice-Activated SOS feature lies advanced voice recognition AI, capable of distinguishing distress signals from regular speech. This technology, seamlessly integrated into wearable devices, empowers users to activate a distress call with a simple voice command. Once activated, the device initiates a rapid communication workflow, transmitting alerts to designated emergency contacts along with real-time location data.

1. INTRODUCTION

In an age defined by technological advancement, the fusion of artificial intelligence (AI) and wearable technology has catalysed remarkable transformations across various facets of our lives. This synergy has ushered in a new era of innovation, one in which devices have evolved from passive tools to active companions, capable of understanding, responding to, and enhancing our daily experiences. At the forefront of this convergence stands the Voice-Activated SOS feature, a pioneering advancement that has redefined personal safety by empowering individuals to swiftly call for help using voice commands.

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Chapter 17

Women's Care With Generalized Metabolism Analysis Report Based on Age Using Deep Learning

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ABSTRACT

Women are facing different problems in different stages. The importance of comprehending women's personal and health issues is highlighted in the opening section. Women are suffering from so many problems, such as peer pressure, depression and anxiety, financial issues, transportation blockades, time management, girls' security and safety, homesickness, career choice perspective, early marriage issues, physical activity and nutrition, substance abuse, mental health issues, physical injury, and violence. The data is grouped by age; for instance, age 8 to 12, age 12 to 16, age 17 to 24, age 25 to 30, age 30 to 40, age 40 to 60, and age 60 to 80. Using deep learning techniques, the proposed system aims to conceptualize and understand these issues, and to develop one website which would provide a solution for each and every query with consulted name and address.

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